

Sprint 4 Intermediate Product Report

Team Name: AI-Driven Robotic Manipulation: Integrating Learning Models with Real-World Control

Team Members: Toan Pham, Albert Vo

Sponsor: Dr. Nakul Gopalan

Date: Nov 4, 2025

1. Overview

Sprint Goal: Optimize cross-embodiment transfer to exceed 50% success rate and conduct comprehensive evaluation

Work Completed:

SBI-401 – Transfer Performance Optimization (In Progress)

2. Achievements

User Stories:

Transfer Enhancement: Transfer Performance Optimization – In Progress

3. Risk Management

Resolved Risks

Risk G: Transfer Performance Gap

Resolution: Achieved 51.2% with optimizations

Key: Curriculum learning + fine-tuning

Risk H: Training Instability

Resolution: Gradient clipping + better initialization

Key: Extensive hyperparameter search

New Risk Emerged

Risk I: Final Capstone Report and Presentation Readiness

Status: Active

Impact: High - capstone final deliverables deadline

Mitigation: Daily writing sprints

Owner: Both

4. Contribution Metrics

Velocity: 42 story points (exceeded 40 planned!)

Task Completion: 41 of 43 tasks (95%)

GPU Hours: 246 hours

Real Robot Time: 2 hours (100% utilized)

Final Report Progress: 5/8 sections drafted

Code Coverage: 68% (up from 45%)

Documentation: 47 pages total

Key Performance Indicators

Primary Goal: Transfer >50% (achieved 51.2%)

Real Robot Demo: Successful

Ablation Study: Comprehensive

Final Report Draft: 65% complete (at risk)

5. Evidence

Taiga Board: [https://tree.taiga.io/project/chickeninvader-](https://tree.taiga.io/project/chickeninvader-chickeninvadercross_model_learning_based_robot_control/us/7?kanban-status=10491383)

[chickeninvadercross_model_learning_based_robot_control/us/7?kanban-status=10491383](https://tree.taiga.io/project/chickeninvader-chickeninvadercross_model_learning_based_robot_control/us/7?kanban-status=10491383)

GitHub Repository: <https://github.com/ChefToan/Cross-Model-Learning-Robot-Control.git>

6. Feedback Summary

Dr. Gopalan: "Exceeding 50% is a significant achievement. The ablation study clearly shows scene graph value. Focus on final report completion and presentation prep for Sprint 5."

Peer Reviewer: "The real robot demo is compelling. Have you considered testing on more diverse tasks?"

7. Updates

Adjustments: No major changes beyond normal sprint refinement.

New or Withdrawn EPICs/User Stories: Updated on the product backlog as needed.

Changes in Priority: Priorities adjusted based on sprint review outcomes.

8. Signature from the Product Owner/Client

Signature/email:

A handwritten signature in red ink, consisting of a large, stylized 'X' or '10' shape followed by a horizontal line.